MD RAKIBUL ISLAM RAIHAN

ML Systems, LLM Inference & Multimodal AI (2027 New Grad)

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Portfolio: theraihanrakibb.github.io

Summary

ML Systems Engineer focused on LLM inference, GPU optimization, and multimodal generative AI. Engineered large-scale LLM serving (Qwen on 4× A800, DeepSeek on H200 clusters), reducing long-context TTFT by 8.3× (55s→6.7s) via 3-tier KV-cache offload and raising stable concurrency by 33% (192→256) via RDMA-based prefill/decode disaggregation. First-author of an IEEE TNNLS manuscript on NVFP4 4-bit video diffusion, benchmarked on H100 GPUs (achieving 4× memory reduction). Strong in SGLang internals, CUDA/Triton, low-precision kernels, and distributed systems.

Experience

AI Infrastructure Engineer Intern *Apr 2026 - Jun 2026*
InfiX.ai, Shenzhen, China

- Reduced long-context TTFT 8.3× (55s→6.7s) and sustained a 92% cache hit rate by designing and deploying a 3-tier KV-cache offload system (GPU HBM → CPU DRAM → SSD) over Mooncake RDMA for Qwen3.5/3.6-27B on 4× A800-80GB (32K+ token contexts).
- Improved stable serving concurrency 33% (192→256 requests) by implementing 2+2 GPU prefill/decode disaggregation with RDMA-based KV transfer, RadixAttention prefix caching, and continuous batching for Qwen3.6-27B.
- Modified SGLang cache-management and memory-allocation internals (*kvcacheio.py*, *memory_pool_host.py*, *hi_mamba_radix_cache.py*) - across v0.5.9 (A800) and v0.5.10-post2 container image (H200) - to resolve production cache failures and support hybrid-attention (Mamba + full-attention) Qwen models.
- Profiled CUDA/Triton inference with Nsight Systems: GEMM accounted for 60% of step time, redirecting optimization from cache I/O to compute kernels; FP8 KV cache halved per-token KV memory (72.9 → 36.5 KB).

Software Engineer Intern (AI Agent) *Jun 2025 - Sep 2025*
Hong Kong Amraim Group Co., Ltd., Shenzhen, China

- Architected and delivered two end-to-end full-stack applications from concept to production deployment, using React front-ends and RESTful back-end services.
- Engineered AI agents that automated key marketing workflows, reducing manual effort and boosting team productivity by 25%.

Electrical Software Engineer Intern *Feb 2025 - Apr 2025*
Shaanxi Longong Intelligent Technology Co., Ltd., Xi'an, China

- Engineered real-time motor-control software on CODESYS PLCs using Modbus/CAN/Ethernet-IP protocols, improving system response time by 15% to IEC 61131 industrial safety standards.

Education

M.Eng. in Software Engineering *Sep 2024 - Jul 2027*
School of Software, Northwestern Polytechnical University
GPA: 88/100, Top 1%, Chinese Government Scholarship, President of International Student Union (ISU)

B.Eng. in Computer Science & Technology *Sep 2020 - Jul 2024*
School of Computer Science, Northwestern Polytechnical University
GPA: 85/100, Top 1%, Presidential Scholarship, Best Undergraduate Thesis Award

Research

Graduate Researcher | Sep 2024 – Present
Audio, Speech and Language Processing Lab (ASLP@NPU)
School of Computer Science, Northwestern Polytechnical University
Advisor: Prof. Zhang Peng

- Developed NVFP4-DiT, a 4-bit audio-guided video diffusion transformer with custom FP4-packed Triton kernels — 3.2× faster than FP16 (2.1× over naive FP4) and 4× memory reduction at competitive generation quality.
- M.Eng. Thesis: "Detecting Deepfake Video by a Multimodal Audio-Visual Framework with Temporal Inconsistencies."

Undergraduate Research Assistant | Jan 2023 – Jul 2024
NWPU Human-Cyber-Physical Fusion Intelligent Computing Innovation Center
School of Computer Science, Northwestern Polytechnical University
Advisor: Dr. Yaxing Chen

- Designed and built a distributed confidential-query framework on Apache Spark + CKKS homomorphic encryption, validated on a 10-node cluster — achieving 95% privacy preservation and a 20% performance improvement via optimized analytics pipelines.
- B.Eng. Thesis: "Design and Implementation of a Distributed Confidential Query Protocol for Apache Spark."

Projects

AI Infrastructure Portfolio, github.com/theraihanrakibb/ai-infra-portfolio
Designed and shipped 10 Dockerized AI-infra services - LLM gateway, model serving, GPU observability, KV-cache simulation, benchmarking, quantization, tracing, and RAG - each with CI and a reproducible quickstart.

Distributed Training Launcher, github.com/theraihanrakibb/train-launcher
Built a PyTorch distributed CLI that launches and supervises multi-node training jobs with fault detection, automatic retry, and checkpoint recovery.

Technical Skills

LLM Inference & Systems: SGLang, vLLM · KV Cache, PD Disaggregation, Continuous Batching, RadixAttention, Mooncake/RDMA
GPU & Performance: CUDA, Triton, GEMM optimization, FP8/FP4/INT4 · Nsight Systems, Nsight Compute, PyTorch Profiler
ML & Multimodal: PyTorch, Hugging Face Transformers · Diffusion/DiT, Multimodal Learning, OpenCV
Distributed Systems: Docker, Kubernetes, Linux, Prometheus, CI/CD
Programming & Fundamentals: Python, C++, CUDA C++, Bash, SQL · Data Structures & Algorithms

Certifications

Deep Learning Specialization · Claude Academy: Claude Code in Action · AI Agents Fundamentals

Publications & Honors

First Author — "NVFP4-DiT: Efficient 4-Bit Audio-Guided Video Diffusion Transformers for Low-Cost Video Generation" — IEEE Transactions on Neural Networks and Learning Systems (TNNLS, Q1), manuscript under review.
Honors: Chinese Government Scholarship (2024-2027); Presidential Scholarship (2020-2024); Wu Yajun Scholarship (2024); Best Undergraduate Thesis Award (2024); Outstanding Graduate (2024), NWPU; Hackathon Winner - NWPU AI Challenge (2023).

MD RAKIBUL ISLAM RAIHAN

ML 系统、*LLM* 推理与多模态 *AI*（*2027* 届应届）

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个人简介

ML 系统工程师，专注大模型推理、GPU 优化与多模态生成式 AI。构建大规模大模型服务（Qwen 部署于 4× A800、DeepSeek 部署于 H200 集群），通过三级 KV-cache 卸载将长上下文 TTFT 降低 8.3×（55s→6.7s），并借助基于 RDMA 的 prefill/decode 分离将稳定并发提升 33%（192→256）。作为第一作者撰写 IEEE TNNLS 论文，研究 NVFP4 4 比特视频扩散，在 H100 GPU 上基准测试并实现显存降低 4×。擅长 SGLang 内部机制、CUDA/Triton、低精度算子与分布式系统。

工作经历

无界智索（深圳）科技（InfiX.ai） *2026.04 – 2026.06*
AI 基础设施工程师实习, 深圳

- 在 4× A800-80GB 上基于 Mooncake RDMA 设计并实现三级 KV-cache 卸载系统（GPU HBM → CPU DRAM → SSD），服务 Qwen3.5/3.6-27B 时将长上下文（32K+ token）首字延迟（TTFT）降低 8.3×（55s→6.7s），缓存命中率维持 92%。
- 通过基于 RDMA 的 KV 传输实现 2+2 GPU prefill/decode 分离，结合 RadixAttention 前缀缓存与连续批处理，将 Qwen3.6-27B 的稳定服务并发提升 33%（192→ 256 请求）。
- 修改 SGLang 缓存管理与内存分配内部模块（*kvcacheio.py*、*memory_pool_host.py*、*hi_mamba_radix_cache.py*）——覆盖 v0.5.9（A800）与 v0.5.10-post2 容器镜像（H200）——修复生产缓存故障并支持混合注意力（Mamba + 全注意力）Qwen 模型。
- 使用 Nsight Systems 分析 CUDA/Triton 推理：GEMM 占单步耗时 60%，由此将优化重心从缓存 I/O 转向计算算子；FP8 KV cache 将单 token KV 显存减半（72.9 → 36.5 KB）。

香港安莱姆集团有限公司（Amraim Group Co. Ltd.） *2025.06 – 2025.09*
软件工程师实习（AI Agent）, 深圳

- 从构想到生产部署，架构并交付两个端到端全栈应用（React 前端 + RESTful 后端服务）。
- 开发 AI Agent 自动化关键营销工作流，减少人工工作量，团队生产力提升 25%。

陕西龙宫智能科技有限公司（Longong） *2025.02 – 2025.04*
电气软件工程师实习, 西安

- 基于 CODESYS 开发 PLC 实时电机控制软件，使用 Modbus、CAN 与 Ethernet/IP 协议，系统响应时间提升 15%，符合 IEC 61131 工业安全标准。

教育背景

软件工程硕士 *2024.09 – 2027.07*
软件学院，西北工业大学（985/211）
GPA: 88/100, 前 1%, 中国政府奖学金, 国际学生联合会主席 (ISU)

计算机科学与技术学士 *2020.09 – 2024.07*
计算机学院，西北工业大学（985/211）
GPA: 85/100, 前 1%, 校长奖学金, 最佳本科毕业论文奖

研究经历

研究生研究员 | 2024.09 – 至今
音频、语音与语言处理研究组（ASLP@NPU）
计算机学院，西北工业大学
导师：张蓬教授

- 开发 NVFP4-DiT：4 比特音频引导视频扩散 Transformer，自研 FP4 打包 Triton 算子——相较 FP16 提速 3.2 倍（相较朴素 FP4 提速 2.1 倍）、显存降低 4 倍，生成质量具有竞争力。
- 硕士论文：《检测深度伪造视频：基于多模态音视频框架与时序不一致性》。

本科研究助理 | 2023.01 – 2024.07
西北工业大学人机物融合智能计算创新中心
计算机学院，西北工业大学
导师：陈亚星博士

- 基于 Apache Spark 与 CKKS 同态加密设计并实现分布式机密查询框架，在 10 节点集群上验证——通过优化分析管线实现 95% 隐私保护与 20% 性能提升。
- 本科论文：《面向 Apache Spark 的分布式机密查询协议设计与实现》。

项目

AI 基础设施代码集, github.com/theraihanrakibb/ai-infra-portfolio
设计并交付 10 个 Docker 化的 AI 基础设施服务——LLM 网关、模型服务、GPU 可观测性、KV-cache 模拟、基准测试、量化、链路追踪与 RAG，均配备 CI 与可复现的上手流程。

分布式训练启动器, github.com/theraihanrakibb/train-launcher
开发 PyTorch 分布式命令行工具，支持多节点训练任务的启动与监控，并具备故障检测、自动重试与断点恢复。

技术技能

大模型推理与系统： SGLang, vLLM · KV Cache, PD 分离, 连续批处理, RadixAttention, Mooncake/RDMA
GPU 与性能： CUDA, Triton, GEMM 优化, FP8/FP4/INT4 · Nsight Systems, Nsight Compute, PyTorch Profiler
机器学习与多模态： PyTorch, Hugging Face Transformers · Diffusion/DiT, 多模态学习, OpenCV
分布式系统： Docker, Kubernetes, Linux, Prometheus, CI/CD
编程与基础： Python, C++, CUDA C++, Bash, SQL · 数据结构与算法

认证与证书

Deep Learning 专项课程 · Claude Academy: Claude Code in Action · AI Agents Fundamentals

论文与荣誉

第一作者 — 《NVFP4-DiT：面向低成本视频生成的高效 4 比特音频引导视频扩散 Transformer》— IEEE TNNLS (Q1)，审稿中。
荣誉：中国政府奖学金（2024-2027）；校长奖学金（2020-2024）；吴亚军奖学金（2024）；最佳本科毕业论文奖（2024）；西北工业大学优秀毕业生（2024）；NWPU AI 挑战赛黑客松冠军（2023）。